

DRY-RUN GUIDE | Linear_search.py
Source: Dsa\search\Linear_search.py

CORE BEHAVIOR

Checks values sequentially until target match or end.

REPRESENTATIVE INPUT

[4,8,2,9], target 2.

STEP TRACE

4 miss; 8 miss; 2 matches at index 2.

EXPECTED OUTPUT / RETURN

Index 2 / source-formatted found result.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Dsa\search\Linear_search.py; line numbers refer to the original file. No source was modified.

```
001 | def linear_search(arr, target):  
002 |     """  
003 |     Performs Linear Search to find the target in the list.  
004 |     Returns the index if found, else returns -1.  
005 |     """  
006 |     for i in range(len(arr)):  
007 |         if arr[i] == target:  
008 |             return i  
009 |     return -1  
010 |  
011 |  
012 | # Example Usage:  
013 | arr = [10, 25, 7, 42, 18]  
014 | target = 42  
015 | result = linear_search(arr, target)  
016 |  
017 | if result != -1:  
018 |     print(f"Element found at index {result}")  
019 | else:  
020 |     print("Element not found")
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.